

AI Hardware & Infrastructure

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Motivation

- AI datacenter is the next “Space Race”

Exploring a space-based,
scalable AI infrastructure
system design

The logo for Project Suncatcher features a stylized sun with wavy rays in shades of yellow and orange, set against a black background.

Project Suncatcher

November 4, 2025 · Travis Reals, Senior Director, Paradigms of Intelligence

**OpenAI and NVIDIA Announce Strategic Partnership
to Deploy 10 Gigawatts of NVIDIA Systems**

Re-Architecting Data-center Lifecycle for AI: A TCO-Driven Framework

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Submitted to ArXiv on 30 Sept 2025

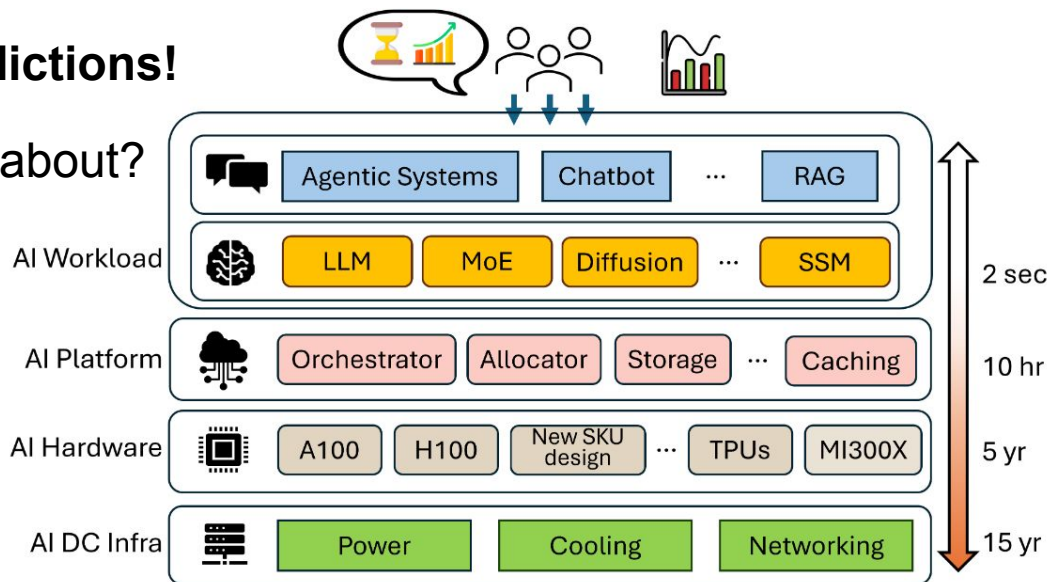
Introduction

What goes into architecting an AI datacenter?

Choices! Trade-offs! Predictions!

What do businesses really care about?

- Raw Performance? 
- Efficiency? 
- Money (Profit) ? 

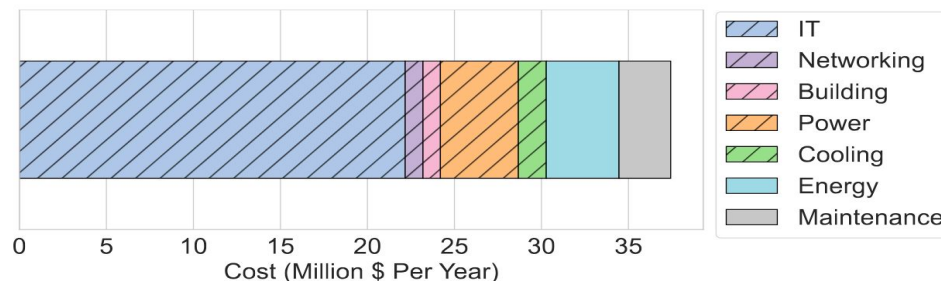


Most optimization efforts have little industrial impact if it does not justify lower TCO

One Metric - Total Cost of Ownership(TCO)

Category	Component	Description	Example Cost (\$)
<i>CapEx</i>	IT	Servers, racks, accelerators, storage	\$375k/server [18]
	Networking	Fabric switches, optics, structured cabling	\$2000/server [59, 85]
	Building	Site preparation, building shell, land, electrical and mechanical base infrastructure	\$0.5/ft ² [29]
	Power	Power infrastructure (Switchgear, transformers, UPS, PDUs, busbars, rack distribution)	\$7.0/W [26, 116]
	Cooling	Cooling infrastructure (chillers, CRAH/CRAC units, pumps, piping, liquid loops, airflow)	\$2.5/W [103, 116]
<i>OpEx</i>	Networking	Port licenses, optics replacement, networking component power	\$600/server [59, 82, 84]
	Energy	IT load scaled by PUE, utility tariffs, demand charges	\$20–40/MWh [35]
	Maintenance	Spares, repairs, monitoring, water/treatment, field-replaceable units, failure-rate	\$5000/server [102]
	Software	Licenses, support contracts	\$200/server

Table 3. TCO components for an example datacenter with DGX H100 [18] servers.



Methodology - Model and Simulate

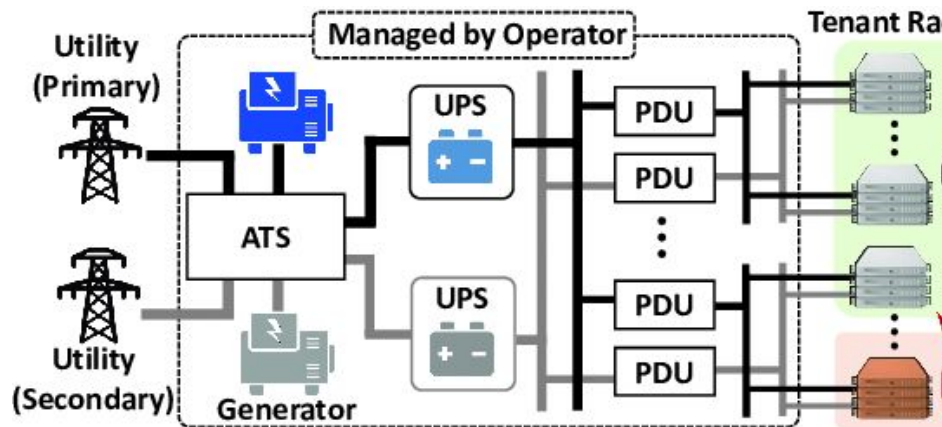
Contribution: First-of-its-kind framework for TCO-based lifecycle management

- **Quantify cost-impact** of intra-stage / inter-stage design choices (data-driven)
 - **Life Cycle:** 15 years (2015 - 2030) with (2015-2025) as validation dataset
 - **Workload:** LLM Inference (DynamoLLM traces), 100k req/s, 15% y.o.y growth
 - **AI Model:** Decoder-only archs, linear size growth (**Other models?**)
 - **Hardware:** Linear growth of FLOPS, Memory Bandwidth, power dissipation and cost
 - **Performance:** Roofline model
- **Fixed SLOs:** 400ms TTFT, 100ms TBT (load increased until SLO violates, to compute max goodput)
- Compute number of servers required until cumulative max goodput reaches projected demand
- Run **Monte-Carlo simulations** to perform state-space search optimizing for TCO

AI data-center lifecycle - Cost-wise Phases

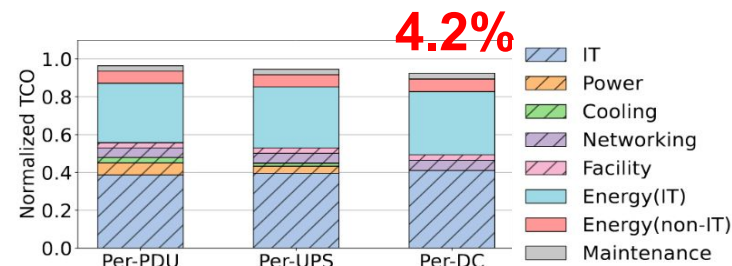
- Building The Infrastructure (Once every **15-30 years**)
 - Power Delivery Infrastructure
 - Cooling Infrastructure
 - Networking Infrastructure
- IT Provisioning (Refresh cycle : Once every **4-6 years**)
- Operation (Day to Day Basis : Once per **inference**)

Impact of Power on TCO



- Each PDU should be provisioned more than peak demand
- **Stranding:** Residual power loss due to lack of distributability
- **Design Complexity and Maintenance**
- **Fault Isolation:** Flatter archs have more risk of total failure

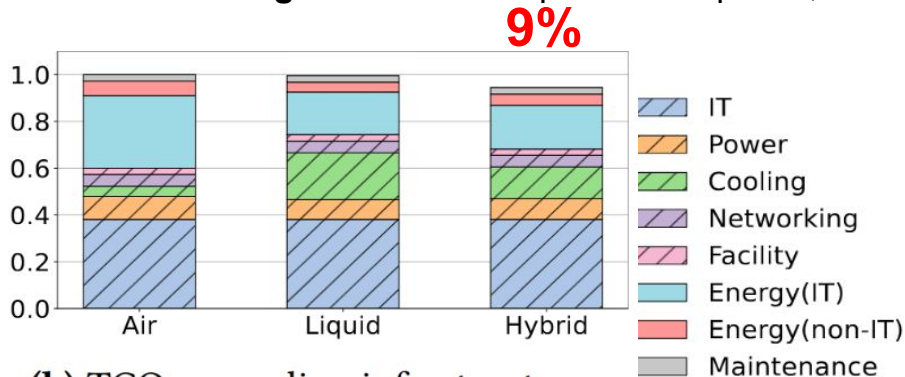
		Power domain		
	Feature	Per-PDU	Per-UPS	Per-DC
CapEx	Stranding	Lower	Medium	Higher
	Complexity	Higher	Medium	Lower
OpEx	Maintenance	Higher	Medium	Lower
Other	Fault isolation	Excellent	Good	Poor



(a) TCO vs. power infrastructure.

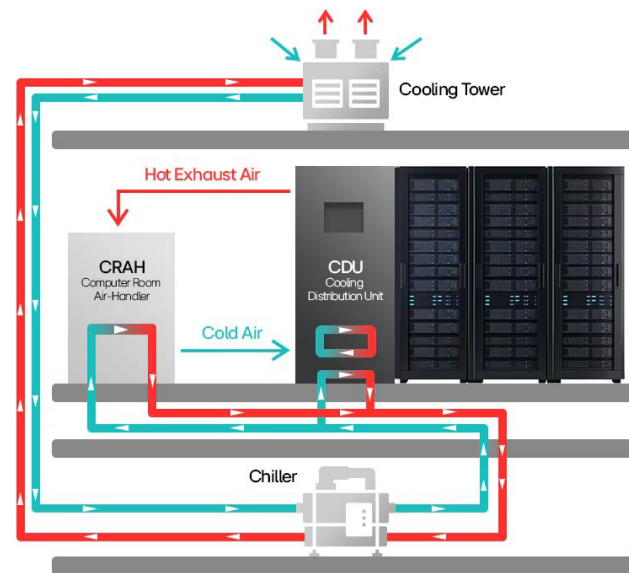
Impact of Cooling on TCO

- **Effective Cooling** helps reduce PUE (Facility Power / IT power)
- **Traditional Cooling:** Fans are used to circulate cool air
- **Liquid Cooling:** Encapsulated by cold Plates/immersed in liquid
- Energy efficiency is reduced through improved heat transfer, reduced chiller load, and lower fan power
- **Ideal Configuration:** 75% liquid-cooled plates, 25% air cooling



(b) TCO vs. cooling infrastructure.

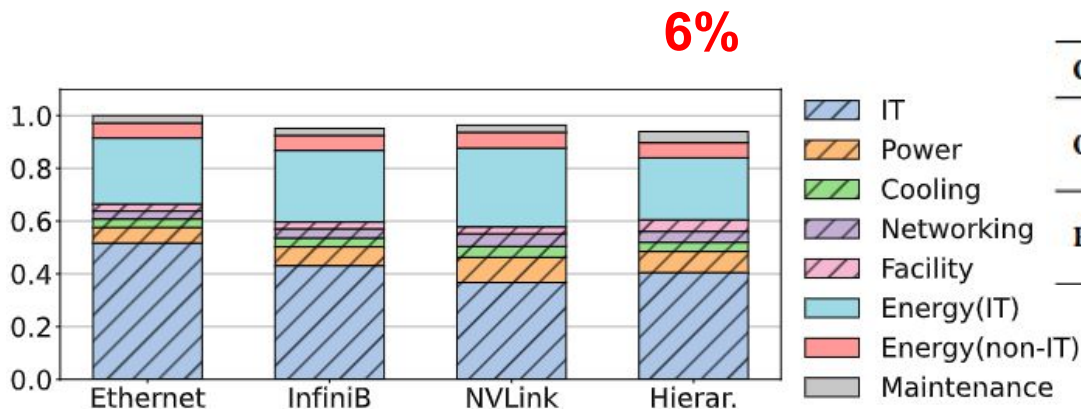
Liquid Cooling in Data Centers



	Feature	Air	Hybrid	Liquid
CapEx	Complexity	Lower	Medium	Higher
	Energy efficiency	Lower	Medium	Higher
OpEx	Maintenance	Lower	Medium	Higher
	High-dense racks	Lower	Medium	Higher
Other	Noise level	Higher	Medium	Lower

Impact of Network on TCO

- **NVLink** is optimal for GPU-dense nodes, high bandwidth (900 GBps), low latency, costly
- **InfiniBand** excels in large-scale low-latency clusters but offers lower peer-to-peer bandwidth (400 Gbps)
- **Ethernet** provides a flexible and cost-effective solution, low bandwidth (400 Gbps), high latency
- **Ideal Configuration:** Hierarchical (NVLink within server, Infiniband within rack, Ethernet across racks)

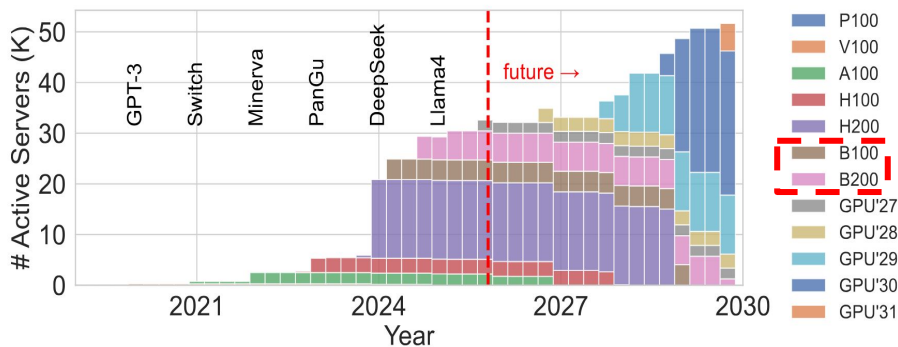


Feature		Ethernet	InfiniBand	NVLink	Hierarchical
CapEx	Cost	Lower	Medium	Higher	Medium
	Energy	Lower	Medium	Higher	Higher
OpEx	Maintain	Lower	Medium	Higher	Medium
	Bandwidth	Lower	Higher	Higher	Higher
Perf	Latency	Higher	Lower	Lower	Medium

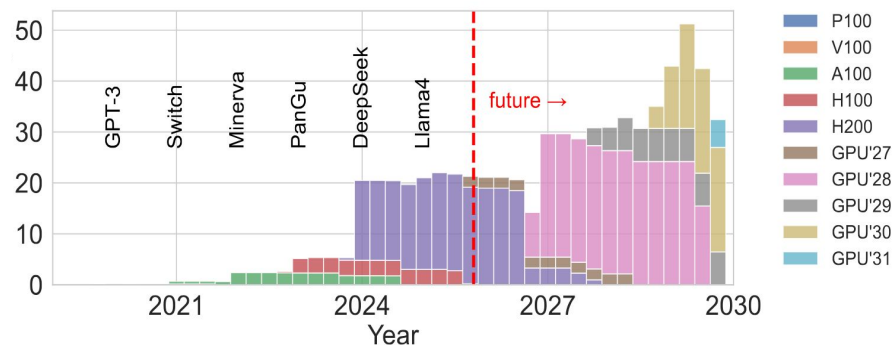
AI data-center lifecycle - Cost-wise Phases

- Building The Infrastructure (Once every **15-30 years**)
- IT Provisioning (Refresh cycle : Once every **4-6 years**)
 - Hardware Refresh (GPU upgrade) : Cost/Performance/Efficiency
 - Model Architecture : Size, Type, Efficiency
- Operation (Day to Day Basis : Once per **inference**)

Impact of Hardware Refresh Policy on TCO



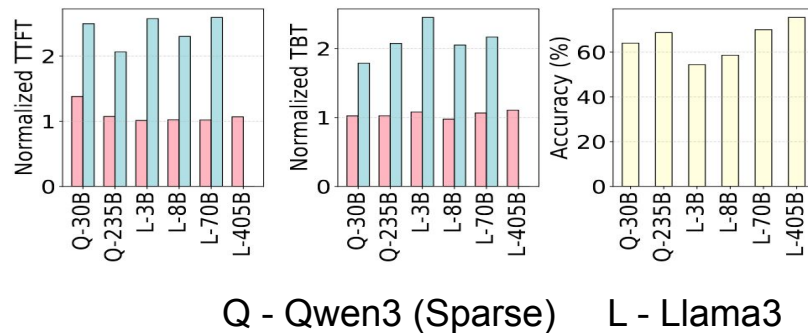
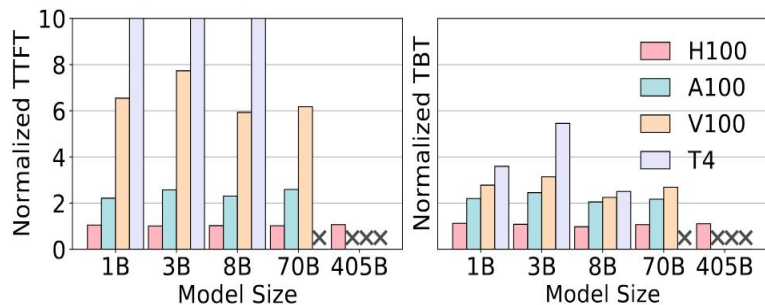
Current Historic Trend Projection



Projection Optimized for TCO

- Aggressively refresh for substantial efficiency gains (e.g., V100 to A100).
- Skip intermediate generations when gains are marginal (e.g. B100/B200)
- Extend life of older hardware if it continues to offer competitive efficiency (e.g. H100/H200)
- Balanced mix of old and new HW offers upto **23%** TCO reduction

Impact of Model Selection on TCO



Q - Qwen3 (Sparse) L - Llama3

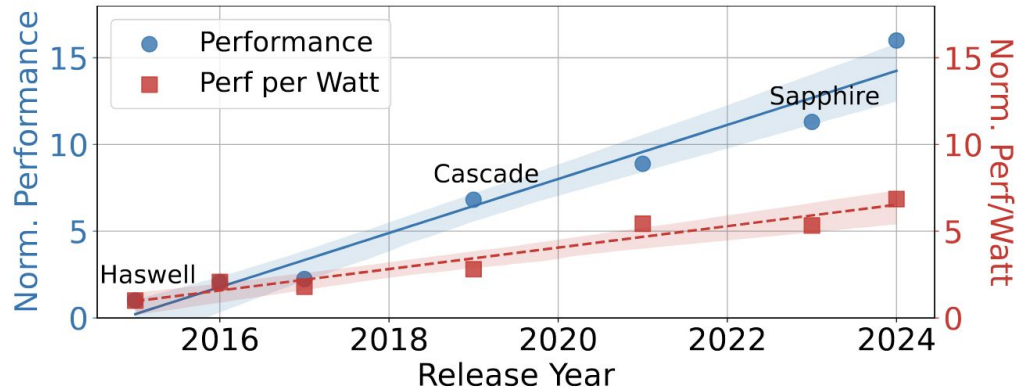
- **Goodput per Watt** degrades on old HW for large models, but not for small models
- Sparse **State-Space Models scale better on older GPUs** compared to Transformer-Based Models
- Sparse models offer **improved latency** but consumes almost **twice memory footprint**
- Increasing model size has lesser impact TTFT/TBT with newer generation of GPUs

AI data-center lifecycle - Cost-wise Phases

- Building The Infrastructure (Once every **15-30 years**)
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Operating an AI Datacenter

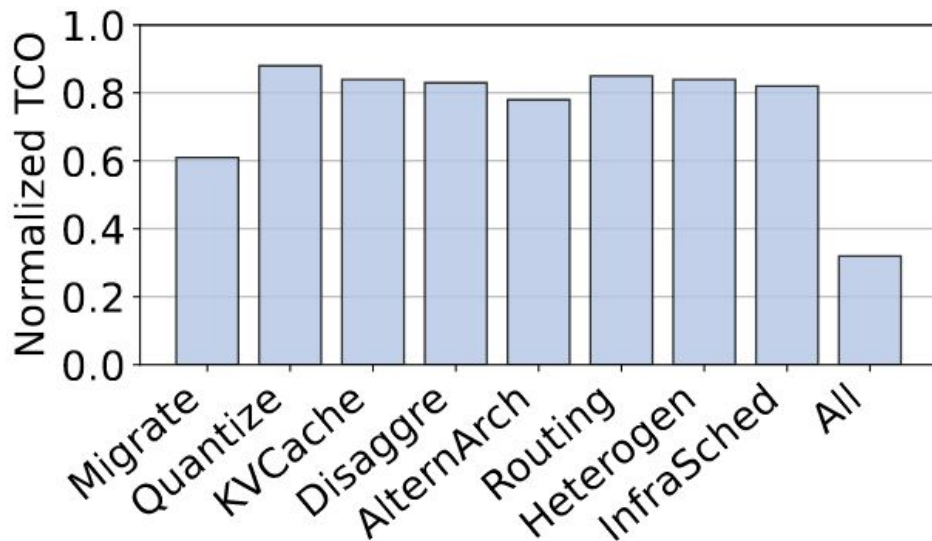
- Sustain high utilization while meeting SLOs across a diverse fleet
- Traditional workload
 - Upgrade to newest generation unless legacy problem
- AI workload
 - No uniform or linear improvements across GPU generations



Operation stage

Optimization Technique	Description	TCO Impact
Smooth Model Migration [45, 80]	Gradual migration from old to newer models upon releases	Avoid rapid hardware procurement
Model Quantization [63, 118]	Lower precision to reduce compute/memory	Lower hardware demand and cost per inference
KV-Cache Management [27, 83]	Optimize storage and reuse of KV cache	Increase older hardware reuse
Disaggregation [83, 89, 105, 120, 122]	Split distinct phases onto different hardware	Extend useful life of heterogeneous generation
Alternative Architectures	Mixture-of-Experts [95], State-Space-Models [38]	Increase older hardware reuse
Model Routing [22, 52]	Direct workloads to the most efficient model variant	Increase older hardware reuse
Heterogeneity-Aware Scheduling [54, 65]	Map workloads to optimal/available hardware generation	Defer refresh costs
Infrastructure-Aware Scheduling [99, 100]	Exploit headroom within infra capacity envelopes	Improve infrastructure efficiency

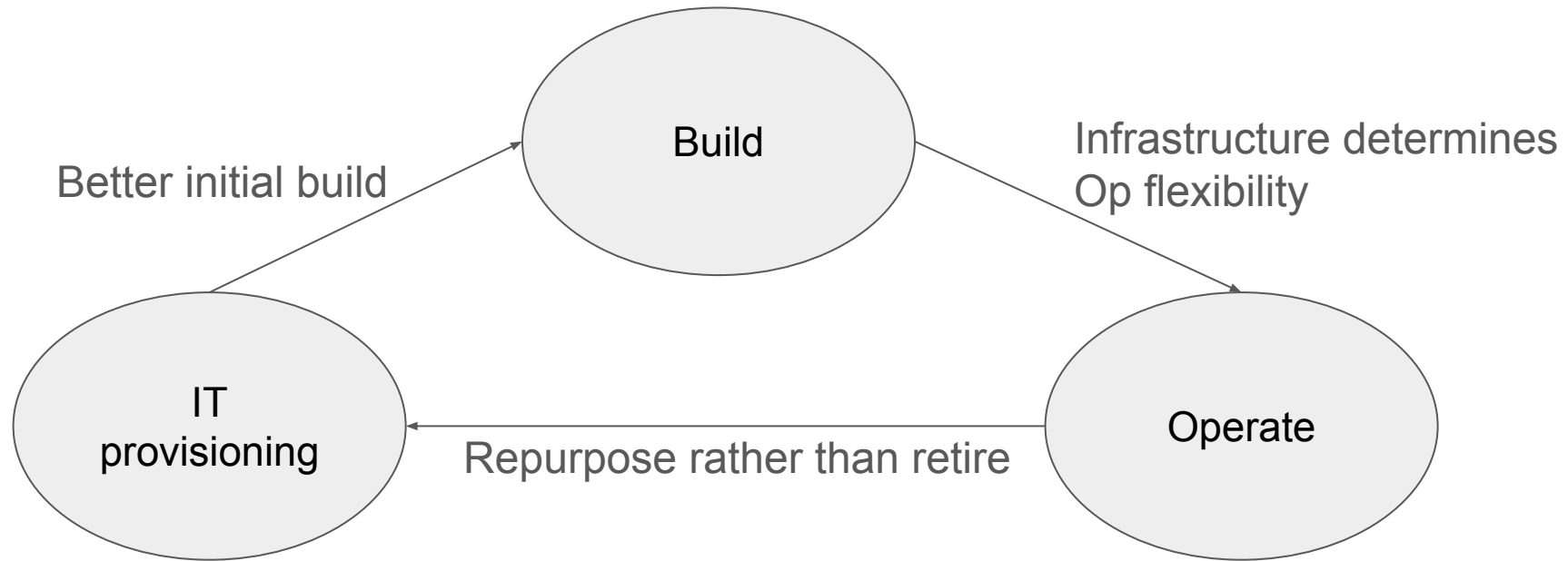
Operation stage



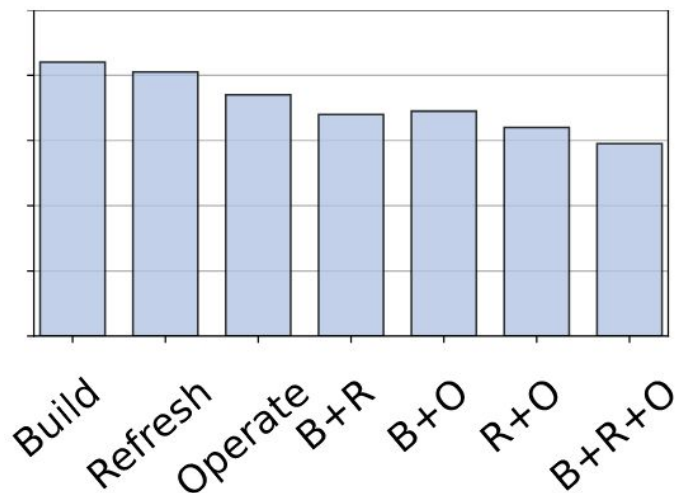
(a) TCO vs. software optimization.

- Smoothen model migration delivers the largest gains by reducing post-release compute load.
- While not fully additive due to overlaps, the cumulative impact is substantial

Cross Stage optimization



Cross Stage optimization



(b) TCO vs. stages.

- Optimizing individual stages yields 20–30% savings
- A holistic approach can reduce TCO by over 40%

Cross Stage optimization

		Model Growth					
		Slow →		Medium ↗		Fast ↑	
Hardware Growth	→	Per-PDU	Air	IB	Per-PDU	Air	NVLink
	↗	Flat	Hybrid	IB	Flat	Hybrid	Hierarchy
	↑	Flat	Liquid	IB	Flat	Hybrid	IB
	→	8 years	Skip		8 years	Skip	
	↗	4 years	Buy new		5 years	Buy new	
	↑	4 years	Buy new		4 years	Buy new	
	→	All Optimizations			Migration + Quantization		
	↗	Disaggregation			All Optimizations		
	↑	Disagg. + Hetero.			Disaggregation		
	→						
	↗						
	↑						

Conclusion

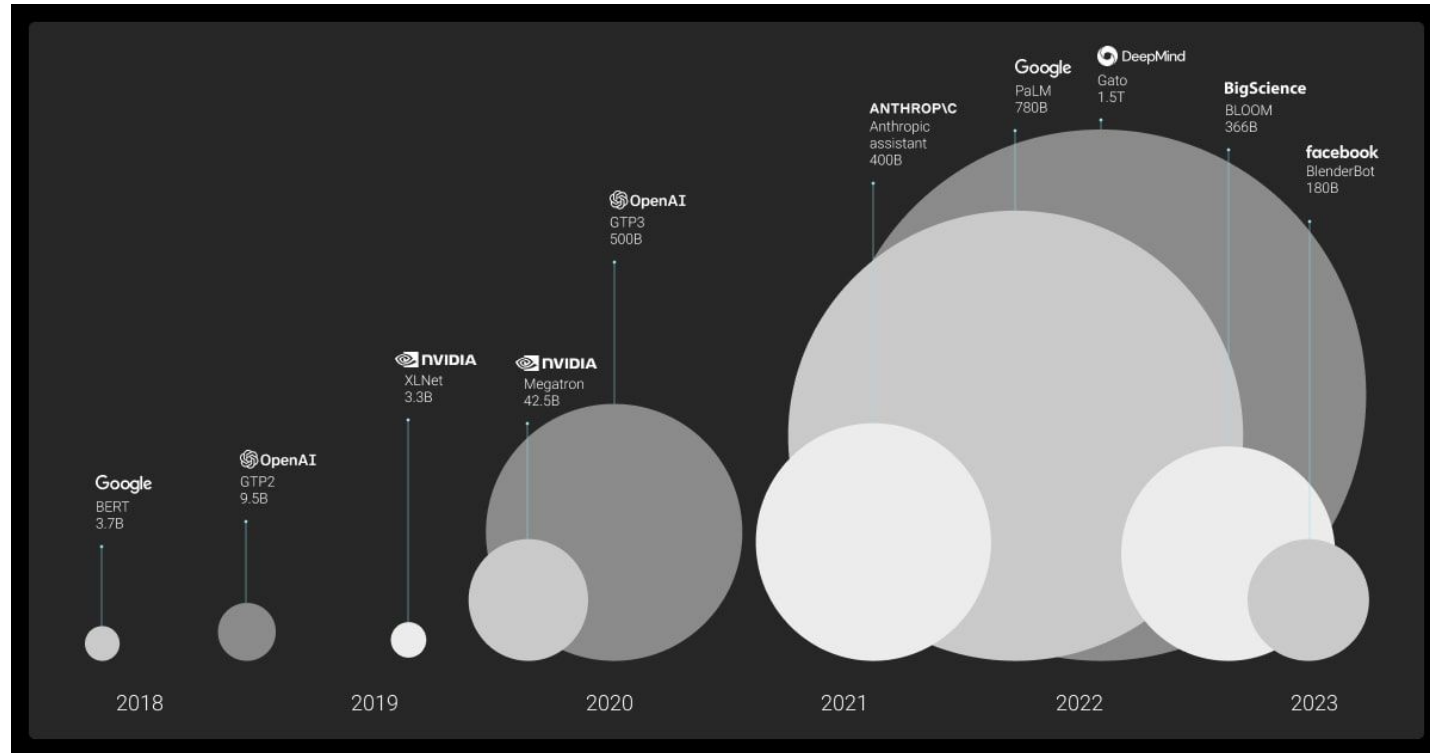
- The rapid growth of AI workloads has outpaced traditional datacenter management.
- While stage-specific improvements (build, IT provisioning, operate) help, the biggest gains, up to 40% TCO reduction, come from a holistic, cross-stage approach

Insights into DeepSeek-V3: Scaling Challenges and Reflections on Hardware for AI Architecture

DeepSeek-AI

ISCA-2025 (Industry Track)

Model Sizes are Growing



Leveling the playing field



DeepSeek-V3 : Uses just 2,048 NVIDIA H800 GPUs, and achieves state-of-the-art performance!

Hardware and Model
co-design!

Outline

- 1.What are the innovations in the V3 model architecture?**
- 2.Why should we care about low-precision computation and communication?**
- 3.Can interconnection driven design be beneficial?**
- 4. Should we focus on large scale network driven design?**
- 5. Current limitations and insights for future hardware architecture design**

Reducing KV Cache size

- In short, the Key-Value cache LLMs remember what was already computed, so they don't have to redo it every time they generate a new token.

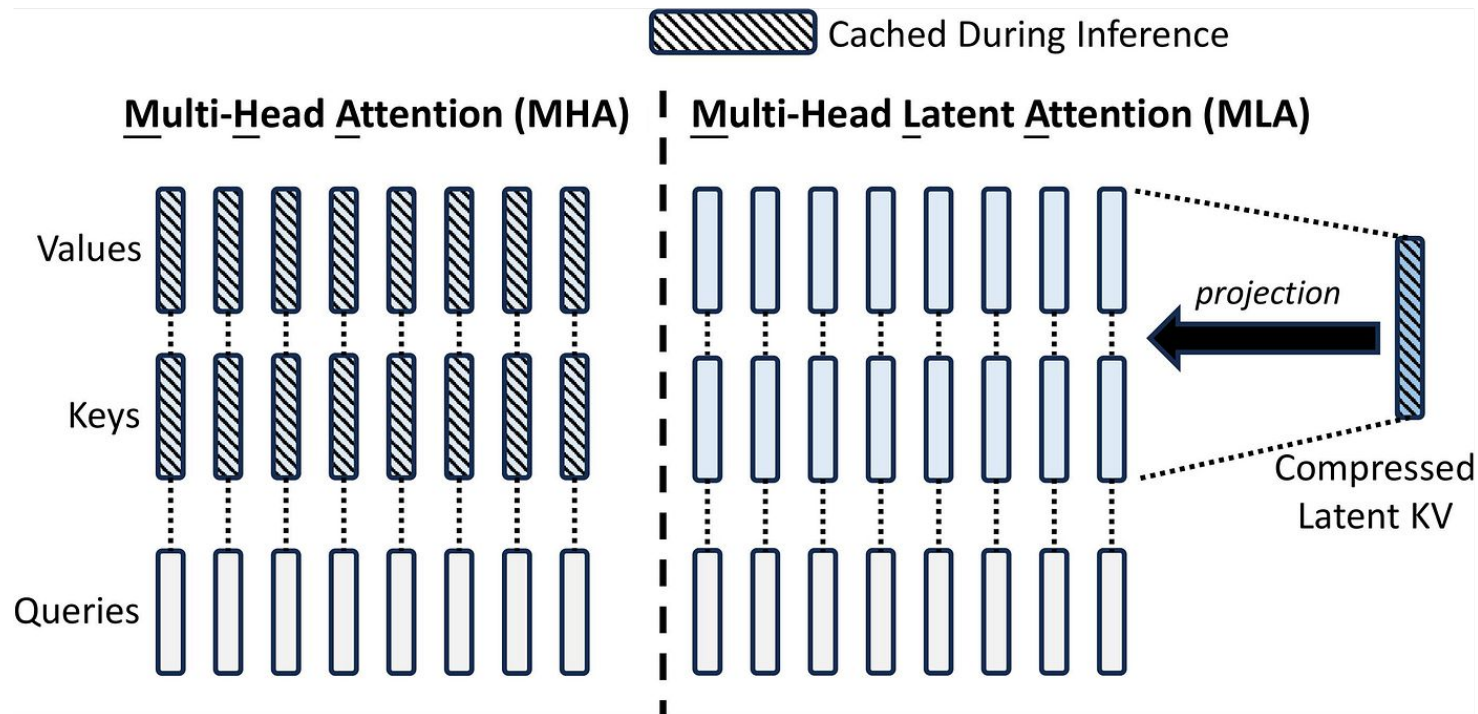
- Limitations of KV caching ✖

- GEMV is much more memory heavy.

- The solution :

- What if we could compress the KV cache so that we could reduce reading data?

Multi-Head Latent Attention (MLA)



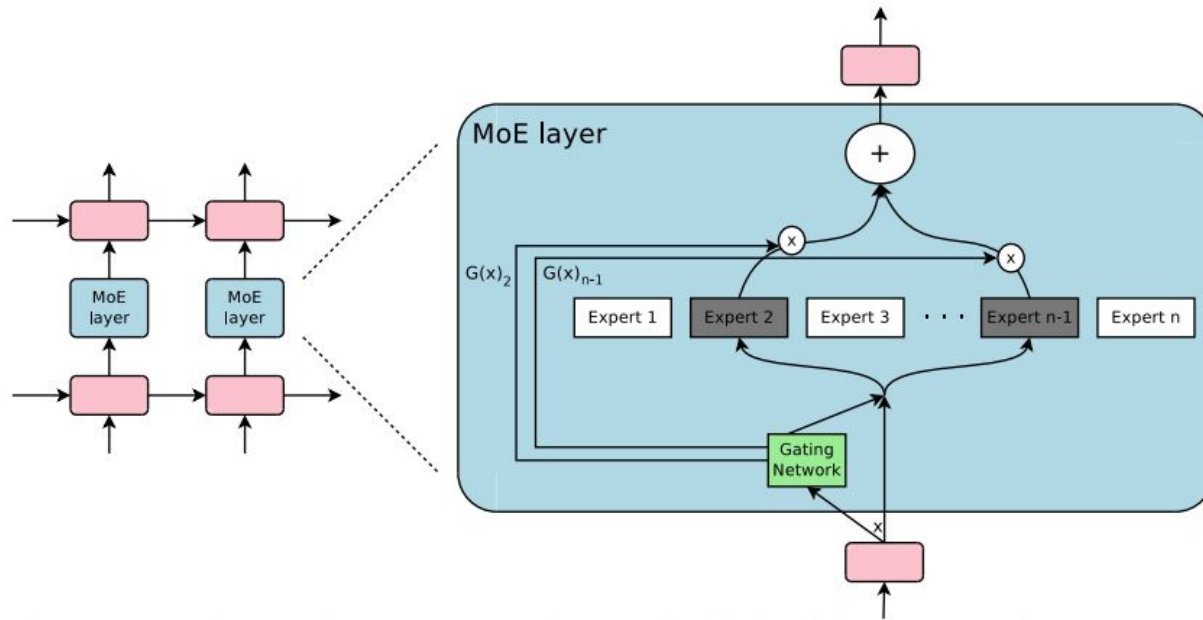
Reductions in KV cache

Model	KV Cache Per Token	Multiplier
DeepSeek-V3 (MLA)	70.272 KB	1x
Qwen-2.5 72B (GQA)	327.680 KB	4.66x
LLaMA-3.1 405B (GQA)	516.096 KB	7.28x

This makes V3 particularly well-suited for scenarios involving long-context processing and resource-constrained environments

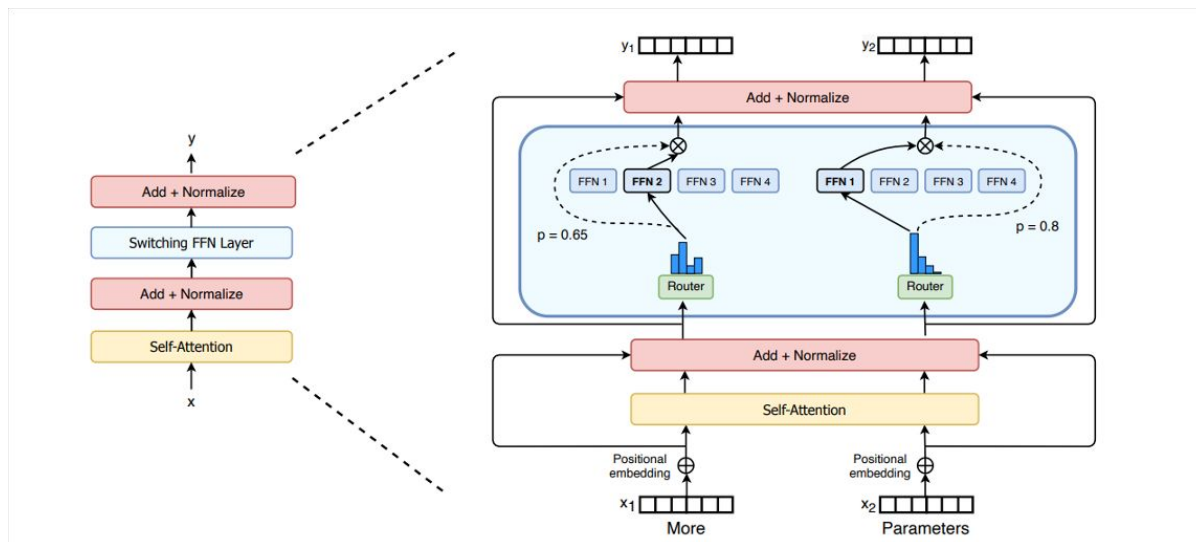
Reducing Computational Requirements for Training

- Mixture of Experts (MoE) selects only a few experts (sub networks).



What's MoE?

- **Sparse MoE layers** are used instead of dense feed-forward network (FFN) layers. MoE layers have a certain number of “experts” (e.g. 8), where each expert is a neural network.
- A **gate network or router**, that determines which tokens are sent to which expert.



Reductions in Computational Requirements for Training

Model	Size	Training Cost
DeepSeek-V2 MoE	236B	155 GFLOPS/Token
DeepSeek-V3 MoE	671B	250 GFLOPS/Token
Qwen-72B Dense	72B	394 GFLOPS/Token
LLaMa-405B Dense	405B	2448 GFLOPS/Token

Increasing Inference Speed

- Inference speed encompasses throughput and latency.
- **Problem:** In MoE, you spend time doing two things:
 1. **Computation** (MLA or MoE math on tokens)
 2. **Communication** (routing tokens between GPUs : dispatch/combine)

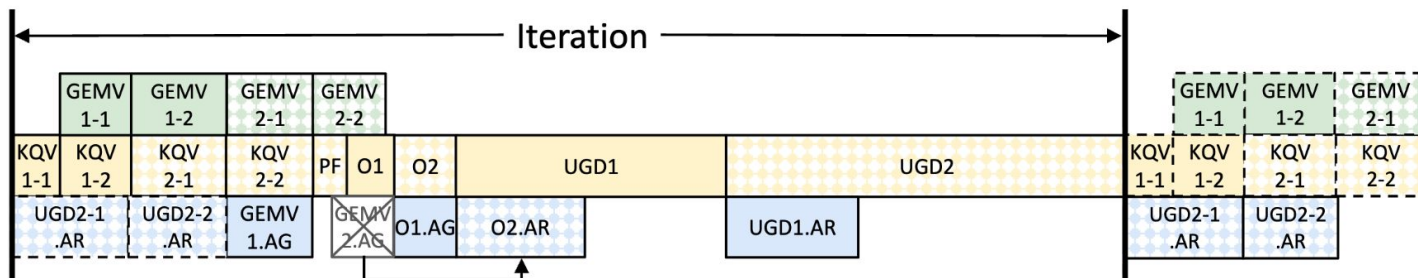
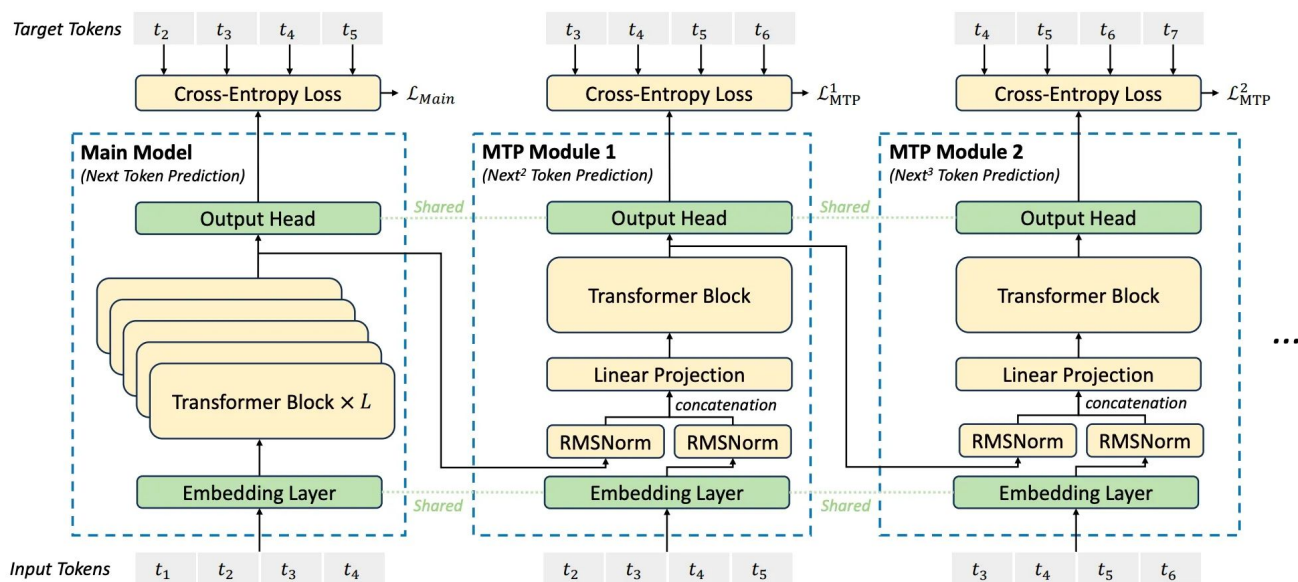


Figure 4. Execution pipeline of NanoFlow. The solid background and shaded background represents operations in two nano-batches that are co-scheduled on the same device. By overlapping the compute-, memory-, and network-intensive operations, NanoFlow increases compute utilization and improves the serving throughput.

Multi-Token Prediction

- During inference, traditional autoregressive models generate one token at a decoding step, leading to sequential bottlenecks.



How Helpful is MTP?

- In DeepSeek-V3, they report **80–90% acceptance rate** for the 2nd token — meaning most predictions were correct.
- That gives about **1.8× faster generation speed (tokens per second)**.

Low-Precision Driven Design

- Previously, there have been models [GPTQ, AWQ] that reduce bit-widths to 8-bit, 4-bit, or even lower, to reduce memory requirements – **in inference**.
- Before DeepSeek-V3, there were no open-source large models leveraging FP8 for training. But why?

Limitations for FP8 in Training

•1. FP8 Accumulation Precision



Increase Precision in Hardware

- Hopper GPUs accumulate these results using **FP22 precision** which is already limited (only 13 mantissa bits).
- This truncation means **tiny differences get lost**, which can cause **instability** during training of large models (gradients can explode or vanish).

•2. Fine-Grained Quantization Challenges



Provide native support for Fine-Grained Quantization

- FP8 can't represent a very wide range of numbers directly, -> **needs scaling factors**.
- These scales can differ per **tile**, **block**, or **tensor**, to avoid overflow or underflow.
- But fine-grained scaling means you must **frequently dequantize and rescale** data

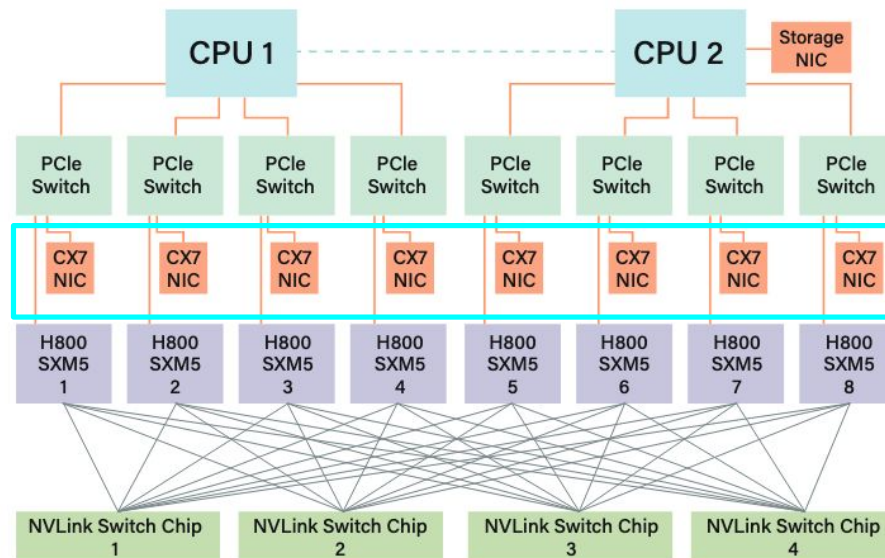
Interconnection Driven Design

Hopper architecture GPUs with reduced

- FP64 computational performance
- NVLink bandwidth 900GB/s -> 400 GB/s

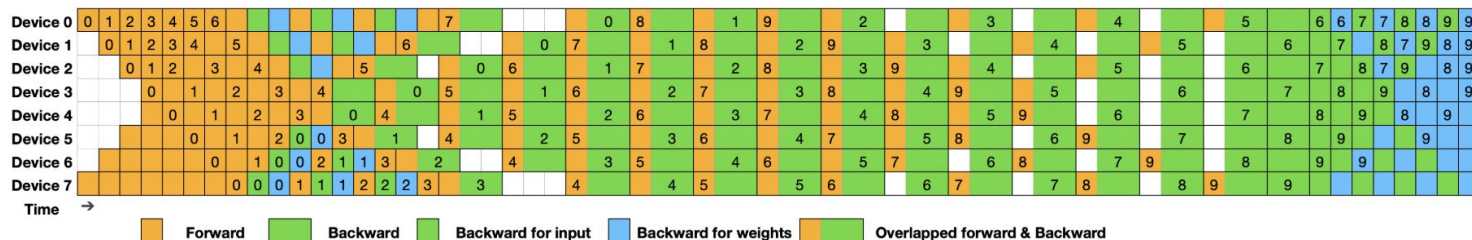
Equip eight Infiniband (IB) NICs (400Gb)

to compensate



Hardware-aware Parallelism

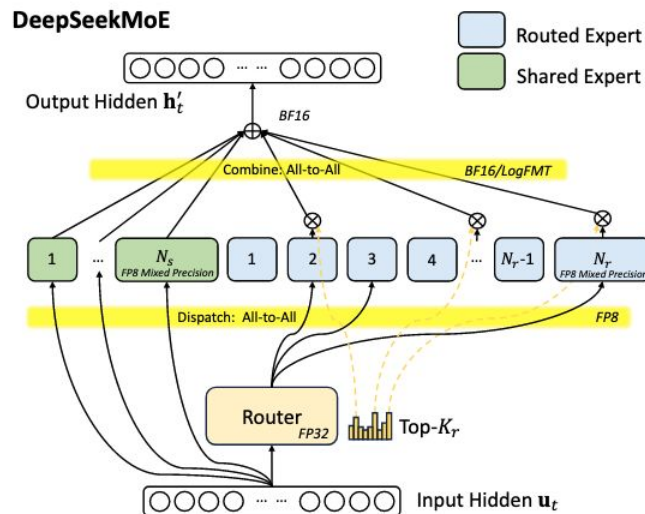
- Tensor Parallelism ❌
- Pipeline Parallelism ✅
- Expert Parallelism ✅



1. DualPipe: DeepSeek-AI. 2025 A bidirectional pipeline parallelism algorithm for computation-communication overlap in V3/R1 training.
2. DeepEP: An efficient expert-parallel communication library

Model Co-Design: Node-Limited Routing

Intra-node 200GB/s > Inter-node 50 GB/s



Algorithmically limit each token routed time inter node (TopK Expert Selection Strategy)

Scale-Up and Scale-Out Convergence

Limitations: Current Routing Strategy reduces **communication bandwidth** requirements, it **complicated communication kernel implementation**

Suggestions:

1. Hardware should integrate intra and inter-node communication into a **unified framework**.
2. **Dedicated Communication Co-Processor**: offload packet processing from GPU SMs
3. **Hardware Synchronization Primitives** : eliminate need for software-based synchronization mechanisms like RDMA completion events

Large Scale Network Driven Design

Multi-Plane Fat-Tree (MPFT) network topology

- Eight GPU and IB NICs for each node
- 1 ethernet RoCE NIC each node

Theoretically supports up to 16384 GPUs.

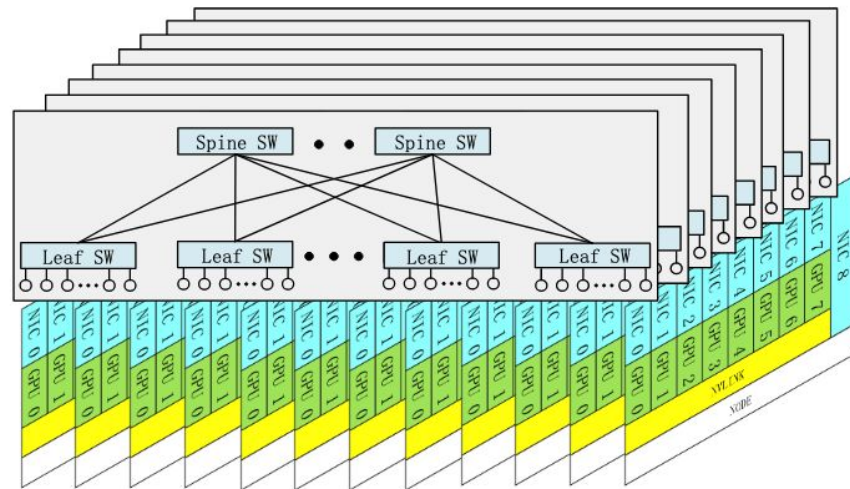
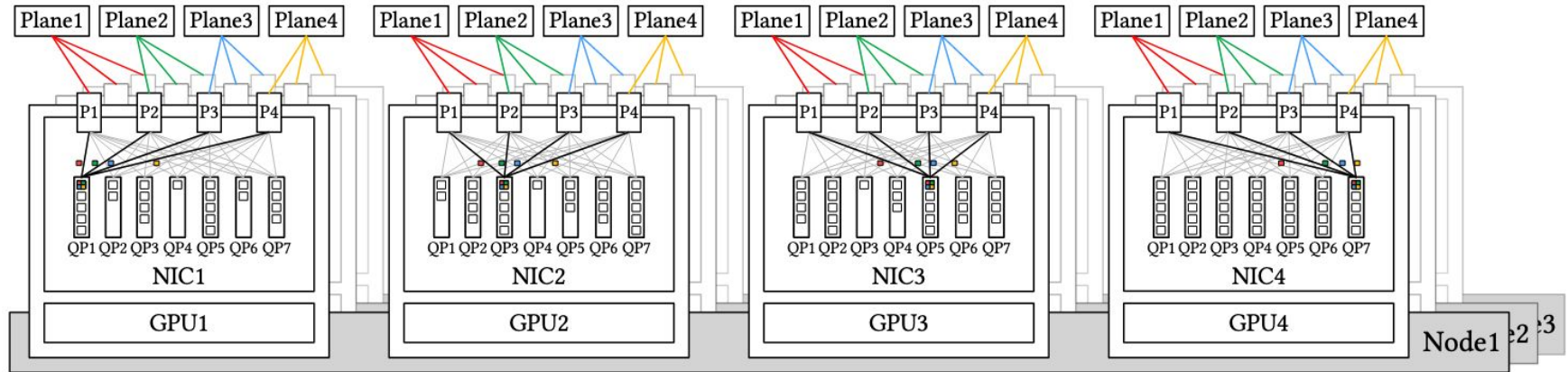


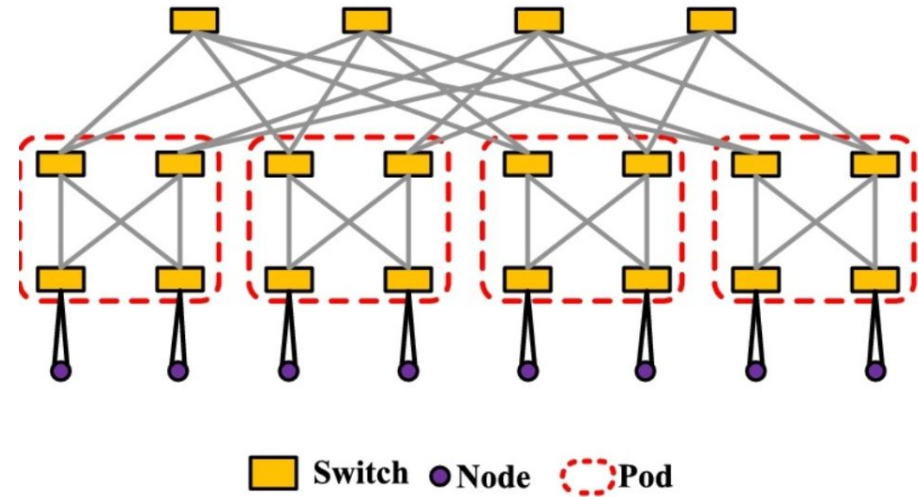
Figure 3: Eight-plane two-layer fat-tree scale-out network: Each GPU and IB NIC pair belongs to one network plane. Cross-plane traffic must use another NIC and PCIe or NVLink for intra-node forwarding.

Large Scale Network Driven Design

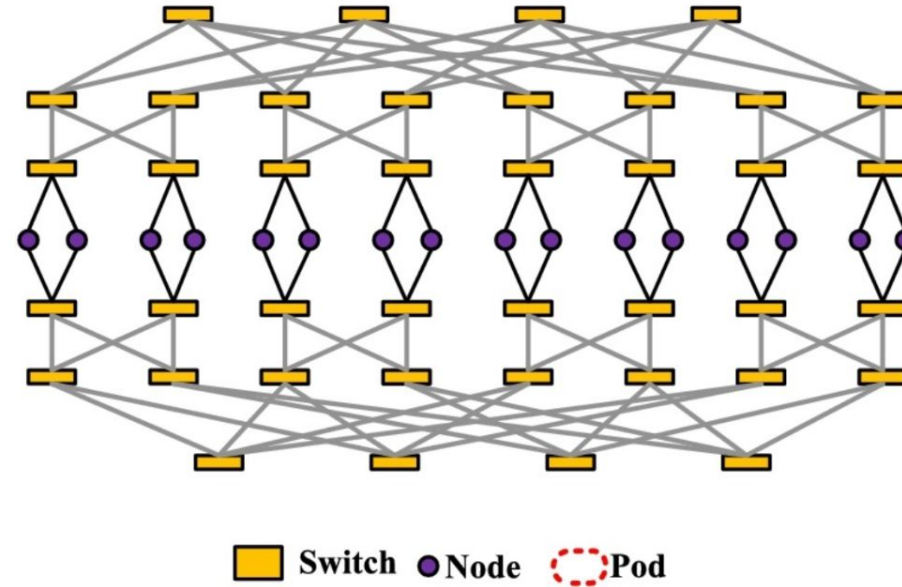
Each QP can simultaneously utilize all available ports



MRFT and MPFT



(c) Multi-Rail Fat-tree.



(d) Multi-Plane Fat-tree

Advantages of MPFT

1. Subset of Multi-Rail Fat-Tree (MRFT): **existing optimization** by NVIDIA and NCCL can be adapted.
2. Cost Efficiency: Fewer Switches and Links overhead *per endpoints*
3. Traffic Isolation: Each Network Plane **operates independently**
4. Robustness: **single-port failures** don't disrupt connectivity and recovery can be rapid and transparent
5. Latency Reduction: Fewer Routing



Metric	FT2	MPFT	FT3	SF	DF
Endpoints	2,048	16,384	65,536	32,928	261,632
Switches	96	768	5,120	1,568	16,352
Links	2,048	16,384	131,072	32,928	384,272
Cost [M\$]	9	72	491	146	1,522
Cost/Endpoint [k\$]	4.39	4.39	7.5	4.4	5.8

Performance compared to MRFT

1. Similar to MRFT attributed to NCCL's

PXN mechanism

2. Each GPU achieves high bandwidth

Over 40GB/s

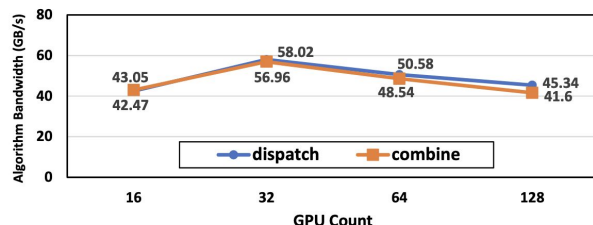


Figure 7: DeepEP performance on MPFT: The EP dispatch and combine kernel communicates across 16 to 128 GPUs using all-to-all. Each GPU processes 4096 tokens. The observed throughput nearly saturates the 400Gps NIC bandwidth.

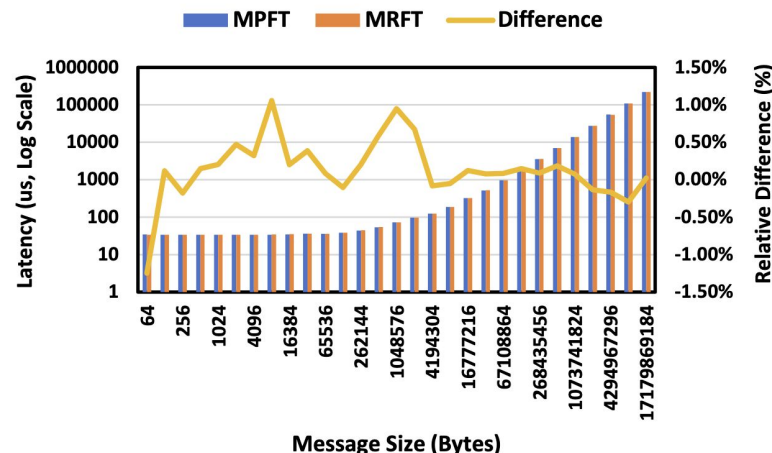


Figure 6: Latency comparison between MPFT and MRFT networks in NCCL all-to-all test under different message sizes, showing that their performance is nearly identical.

Robustness Challenges

Limitations:

1. Interconnect Failures
2. Single Hardware Failures (e.g node crashes, GPU failures, ECC memory errors)
3. Silent data Corruption (rely on application-level heuristics)

Suggestions:

1. Advanced Error Detections (checksum-based validation) on hardware
2. Comprehensive diagnostic toolkit should be provided by hardware vendors.

Conclusion

- Exemplifies the potential of hardware-software co-design in scalability robustness and efficiency.
- Propose actionable recommendations and roadmap for next-gen AI-optimized hardware